



The Future of Work

AI, Automation & the New Workforce

Dr. Pinaki Bhaskar · AI Research Architect · Samsung R&D Institute India

Distinguished Lecture · BIM 43 Induction · BIM Tiruchirappalli · 2 July 2026

Dr. Pinaki Bhaskar

Current Role

AI Research Architect & Engineering Manager

Visual GenAI Group, Samsung R&D Institute India · Since 2015

Research Focus

Generative AI · VLM · LLM · Knowledge Graphs · NLP

Education

Ph.D. Computer Science & Engineering, Jadavpur University

Postdoc · CNR-IIT, Pisa, Italy (ERCIM Fellowship)

Key Achievements

8

Global Patents

36

Publications

11+

Years Leading ML

3×

Best Project Award

Ships AI to Millions

Led VLM-based Semantic Search shipped in **Galaxy S24 & S25** series.
On-device Knowledge Graph commercialized from **Galaxy S21** onwards.

Recognition

Zinnov "Technical Role Model – AI & BDA" (2022) · Board of Studies,
NIT Meghalaya · Samsung ACE Employee

60-Minute Journey

Agenda



The Shifting Landscape

AI, automation & megatrends reshaping work · ~10 min



AI by the Numbers

What data tells us about jobs, skills & wages · ~10 min



Inside the AI Lab

Real-world AI development at Samsung · ~12 min



The Two-Track Labour Market

Who wins, who adapts, what roles thrive · ~8 min



Skills for the Future Manager

Human skills + AI literacy = leadership edge · ~10 min



Your MBA Playbook

Strategies for leading in AI-augmented orgs · ~10 min

Part 1

The Shifting Landscape

Five megatrends converging to redefine what work means — and what managers must become.

Five Megatrends Reshaping Work



AI & Automation

GenAI reshaping tasks in every sector



Digital Ecosystems

Cloud, IoT & platform economies



Workforce Shifts

Hybrid work, gig economy, Gen Z



Demographic Change

Aging workforce + emerging market talent



Sustainability Mandate

ESG, green tech & purpose-driven orgs

None of these forces act alone. Their **intersection** is where future managers will live and lead every day.

Part 2

AI by the Numbers

The data is clear — AI isn't just a tech story. It's an economic, social, and human story.

Key Findings



The Two-Track Labour Market



Track A: "Professionalised" Roles

AI *augments* these roles — amplifying human expertise and demanding higher-order skills

- Growing **2x faster** than democratised roles
- **42% higher** wage growth since 2021
- Demand: judgement, leadership, creativity
- Examples: AI Architects, Strategy Analysts, Product Managers, Research Scientists



Track B: "Democratised" Roles

AI *simplifies* these tasks — reducing skill barriers but also reducing value differentiation

- Tasks become easier for non-experts
- Lower wage premiums and slower growth
- Routine: templated writing, basic data entry, tier-1 support
- Risk: commoditisation over time

The question for your MBA class is not "Will AI replace me?" but "Which track am I building towards?"

Part 3

Inside the AI Lab

How a real-world AI R&D team ships products to hundreds of millions of users —
lessons for future leaders.

How AI Products Are Actually Built



Research & Experimentation

Literature survey, model architecture design, dataset curation, training experiments & evaluation metrics



Optimise for Device

Quantisation (INT4/INT8), pruning, on-device NPU deployment — making billion-param models run on a 6g phone



Cross-Geography Collab

Teams across Seoul (SR), Suwon (MX), Bangalore (SRI-B) — aligning research cadence with product timelines



People Leadership

Managing multi-disciplinary ML teams; mentoring PhDs & engineers; balancing innovation with delivery pressure



Strategy & Market Intel

Tracking competitor moves (Apple, Google, OpenAI) to shape Samsung's 2–5 year AI roadmap



From Research to Product

From research paper to 100M+ device feature: rigorous QA, privacy compliance & OTA update pipelines

Part 4 & 5

Skills for the Future Manager

The skills that will define whether you lead in an AI-augmented world — or get left behind by it.

The T-Shaped Leader of 2030

 Human Skills (Irreplaceable)



Human Judgment
Deciding when AI is wrong. Knowing what to question.



Empathy & Stakeholder Management
2.5× more valuable in AI-exposed roles (PwC, 2026)



Creativity & Innovation
Framing problems AI can't frame itself



Ethics & Responsible Leadership
Deploying AI that is fair, safe & trustworthy

 AI Literacy (Non-Negotiable)



Understanding AI Capabilities & Limits
What GenAI, LLMs, VLMs can & cannot do



AI Collaboration
Getting the best from AI tools as a co-worker



Data Fluency
Reading dashboards, questioning models, building intuition



Continuous Learning Mindset
Skills changing 2× faster — learn-unlearn-relearn daily

"AI Will Reshape More Jobs Than It Replaces"

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- AI/ML Specialists & Engineers
- Data Analysts & Scientists
- AI Ethics Officers
- Human-AI Interaction Designers
- Green Energy & Climate Tech Roles
- Care Economy (Healthcare, Education)

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- Routine data entry & processing clerks
- Basic customer service agents
- Repetitive assembly-line roles
- Standard bookkeeping & payroll
- First-tier tech support scripted roles
- Template-driven report writers

WEF projects **170 million new jobs** created vs **92 million displaced** → net gain of **78 million jobs globally by 2030**. The opportunity is real — but it requires deliberate skill-building.

THE FUTURE OF WORK

Position for Track A

Seek roles that demand judgement, strategy & human expertise. Avoid commoditised task work. Ask: "Does this role grow with AI or get replaced by it?"

Build a Dual-Competence Stack

Domain expertise (marketing, finance, ops) PLUS AI fluency = rare & highly valued combination. Neither alone is enough anymore.

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Bias, privacy & fairness are management problems, not just tech problems. MBA leaders who understand AI ethics will be trusted to deploy it at scale.

Adopt AI as a Co-Worker

Daily GenAI users report higher pay, job security & productivity (PwC 2025). Integrate AI tools into every workflow — but own the decision-making.

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Future orgs are flat, distributed & multi-disciplinary. Bridge technical AI teams with business functions — be the translator, not just the specialist.

Embrace Lifelong Learning

Skills obsolescence is accelerating. The most valuable trait of 2030: the ability to rapidly acquire new competencies. Your MBA is the beginning, not the peak.

Three Questions for BIM 43 to Carry

01

How do I responsibly handle information processing — maximizing value without creating harm or dependency?
AI handles information processing. Leaders handle meaning, values, and consequence.

02

How do I responsibly deploy AI in my organization — maximizing value without creating harm or creating harm or dependency?
The challenge is responsible deployment beyond ROI to societal impact.

03

Am I building a career based on timeless human strengths — or tasks that AI will commoditise within 5 years?
Reskill before you're asked to. The best time to adapt is before disruption, not after.

THE FUTURE

- ◆ **AI is a job transformer, not a job terminator.** 170M new roles emerge; 92M change. Net positive — if you adapt.

- ◆ **AI will create more jobs than it destroys.** Double down on human strengths.

- ◆ **Junior roles now demand senior skills.** Leadership & strategic thinking are day-one requirements. Step up early.

- ◆ **Two-track labour market is already here.** Professionalised roles: 42% higher wages, 2x growth. Choose your track deliberately.

- ◆ **AI adoption is accelerating.** Early adopters win — personally and organizationally.

- ◆ **Domain + AI fluency = career edge.** Neither alone is sufficient. BIM 43 has the rare chance to build both from day one.



Thank You, BIM 43

The future of work is not something that happens *to* you. It is something you shape
— with skill, with courage, and with humanity.

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Distinguished Lecture

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